

|   |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
|---|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 2 | (a)  | <p>pressure = force / area OR weight (of liquid) / area [B1]</p> $p = \frac{\text{mass} \times g}{A} = \frac{\text{density} \times \text{volume} \times g}{A} = \frac{\rho Ahg}{A}$ <p>[M1]</p> <p>[A0]</p> <p><i>deduct 1 mark if any symbol used is not defined except for <math>\rho</math>, A, g and h</i></p> <p><i>no marks if concept of upthrust used</i></p>                                                                                                                                                                        |  |
|   | (b)  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
|   | (i)  | Total pressure includes atmospheric pressure. [B1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
|   | (ii) | <p><math>P = h\rho g + P_{atm}</math></p> <p>Gradient of the graph is <math>\rho g</math> [C1]</p> <p>Gradient = <math>\frac{(3.0-1.0) \times 10^5}{1.5-0} = 133333</math></p> <p><math>\rho = \frac{133333}{9.81}</math> [M1]</p> <p><math>= 13600 \text{ kg m}^{-3} \approx 14000 \text{ kg m}^{-3}</math> [A1]</p>                                                                                                                                                                                                                        |  |
|   | (c)  | <p>Upthrust is given by <math>V\rho_l g</math> where <math>V</math> is volume of the object and <math>\rho_l</math> the density of the fluid</p> <p>The weight of the object is given by <math>V\rho_o g</math> where <math>\rho_o</math> is the density of the object. [B1]</p> <p>To experience an <u>upwards net force</u> when the object is fully immersed such that it floats, <u>the upthrust is greater than the weight</u>. [M1]</p> <p>(or when object floats, weight = upthrust B1, water displaced &lt; volume of object M1)</p> |  |

2

3